

THE FOUR RECORDS, AND WHICH CAN CHANGE



Decision stores the probability, risk level, the rules' signals, the full feature snapshot and the versions of model, ruleset, thresholds and feature spec, so any decision can be reproduced months later. **Case** holds many alerts: a new alert joins an **open** case with the same `subject_token` (the customer, not the account) within **24 hours**; otherwise it opens a new case. Since spec 1.2.0 a payment is one type of **event** (D72): logins, device binding, credential and SIM changes, payee additions and credits share one envelope, are tokenised at the same boundary, and are read by measurements; only payments and credits are decided.

TRANSACTION EVENT — FIELDS

FIELD	DEFINITION
transaction_ref	Sender's reference; the idempotency key. A repeat is a duplicate
occurred_at	When it happened. Every behavioural measurement uses this
ingested_at	When we received it. Never used for measurements
amount_minor	Amount in kobo , as an integer; never floating point
currency	ISO 4217 code (NGN)
channel	MOBILE_APP · WEB · USSD · POS · ATM · AGENT · BRANCH · API
instrument	CARD · ACCOUNT_TRANSFER · CASH · WALLET
rail	NIP · NEFT · RTGS · CARD_SCHEME · INTRABANK · ATM_NETWORK
auth_result	APPROVED · DECLINED · FAILED · REVERSED — an attempt
decline_reason	Coarse category, e.g. INSUFFICIENT_FUNDS; else empty
subject_token	One-way hash of the customer : what a case is about
account_token	One-way hash of our customer's account : what a baseline is about
direction	OUTBOUND (a payment) · INBOUND (a credit, D78)
remitter_token	Credits only: hashed sending account, with its bank code
bvn_token	Hashed BVN, stamped at intake from the core system (D75); keys the industry watch-list
beneficiary_token	One-way hash of the receiving account; empty for card, cash
device_token	Hashed device fingerprint
ip_region	Coarse region, never a raw IP address
merchant_category	Card and POS only
display_name	Obviously invented name, analyst screen only
account_opened_at	Account fact stamped at intake , never changed afterwards
last_activity_at	Stamped fact; dormancy is worked out from it
product_type	SAVINGS · CURRENT · DOMICILIARY · WALLET
origin_sol_id	Branch that owns the account; coarse geography

CASE STATES

OPEN → UNDER_REVIEW → CONFIRMED_FRAUD | FALSE_POSITIVE | INCONCLUSIVE → CLOSED
UNDER_REVIEW → ESCALATED → INFOSEC | FRAUD_OPS → back to review, or CLOSED

Inconclusive is mandatory: a forced yes/no under time pressure poisons the labels. Outcomes are the only real training labels this system will ever produce.

DECISIONS THAT SHAPE THE DATA

D9b	Two timestamps	Late batches scored on arrival time fake a burst; keeping one timestamp loses the truth for good.
—	Integer kobo	Floating-point money is finance software's commonest defect; every total inherits it.
D19	Two identity levels	Cases follow the customer, so one takeover across three accounts is one case; baselines follow the account.
D9c–d	One-way tokens	Hashed with a secret pepper at intake; no reversal table. The token travels; the receiving bank looks it up.
D21	Attempts, not successes	Without refusals, card testing is invisible and reversals count money twice.
D22	Stamp, never join	Joining today's account table would leak the future into training and change old decisions.
D15	One measurement package	Training and live call the same functions; a test proves identical numbers.
D25, D77–78	Twelve, then twenty-six	Twelve cut from fifteen to pay for the identity work; five event and five credit measurements once their data existed; four for the Nigerian fraud types (D82).
D78b	The model reads twenty-one	The five receiving-side measurements made unseen takeover worse (0.991 to 0.857); rules read them, every decision stores all.
D64	–1 and the cap	The old marker 999 read as a trusted old payee and helped hide card-testing fraud.
D10f	No borrowed scores	Never train on another party's "risk", "score", "rate" or "fraud" column unless its code can be read.

THE TWENTY-SIX MEASUREMENTS TAKEN ON EVERY PAYMENT AND CREDIT (FEATURE SPEC 1.4.1)

THE PAYMENT AND THE ACCOUNT · D25, D64		BEFORE THE MONEY MOVED · EVENTS · D77	
amount_log10	Size of the payment, log scale	failed_logins_1h_subject	Failed logins for the customer, last hour
amount_ratio_to_account_p95_30d	÷ the account's largest usual payment	device_bound_hours	Hours since this device was bound; capped at 72
txn_count_1h_account	Attempts on the account, last hour	credential_changed_hours	Hours since a PIN or password change; capped at 72
approved_value_ratio_24h_vs_daily_mean_30d	Money out today ÷ a normal day	sim_changed_hours	Hours since a SIM swap; capped at 72
failed_attempts_1h_account	Refused or failed, last hour	payee_added_minutes	Minutes since this destination was enrolled
decline_rate_24h_account	Share of today's attempts refused	MONEY COMING IN · CREDITS · D78, RULES ONLY	
distinct_beneficiaries_1h_account	Destinations paid, last hour	credits_24h_account	Credits into the account, last day
beneficiary_is_new_to_account	1 if never paid by this account	distinct_remitters_24h_account	Different senders, last day
beneficiary_first_seen_days	Days since the bank first saw it (0 = new)	inbound_count_ratio_24h_vs_daily_mean_30d	Credits today ÷ a normal day
device_is_new_to_subject	1 if the customer never used it	minutes_since_last_credit	Since money last arrived; capped at a day
account_age_days	Days since opening, from occurred_at	pass_through_ratio_24h	Money out ÷ money in, last day
days_since_account_activity	Days the account had been quiet		
WHERE, HOW AND WHEN · D82			
region_is_new_to_subject	1 if the customer had not used this region this month, before today		
card_present_count_1h_account	Card uses at POS, ATM or agent in the hour		
beneficiary_distinct_senders_24h	Other customers who paid this destination today		
hour_of_day_local	Hour in Nigeria, 0 to 23		

-1 means "does not apply" (no destination, no date) and sits outside every real range. **Ratios are capped at 100.** History looked at: account 30 days, customer 30 days, destinations 90 days, events 72 hours, credits 30 days. Outgoing history reads OUTBOUND only, so no older measurement changed meaning when credits arrived. The spec version is stored on every decision.

Sources: PRD §6 and §10, backend/riskradar/features/spec.py, docs/decisions.md. Synthetic data only; before any real data, an NDPA 2023 lawful-basis assessment and retention schedule are required.